

3.2 Logarithmic and exponential functions

Pure Mathematics 3 · 2026 examination · Concept guide and original exam-style practice

Understand the concept

Use logarithms to linearise exponential equations and transform data when a model $y=Ae^{(kx)}$ is justified. Differentiation $d(e^{(kx)})/dx=ke^{(kx)}$; $d(\ln|abs(x)|)/dx=1/x$ for $x \neq 0$. Check original domain after solving logarithmic expressions.

Worked example: follow the reasoning

If $y=3e^{(2x)}$, then $\ln y = \ln 3 + 2x$ when $y > 0$, so log plot gradient is 2.

Structured practice

Answer in the space provided. Show working and units where relevant.

1 Solve $e^{(2x)}=7$ and give exact x .

[2]

Final answer: _____

Answers and explanation

Question 1 [2 marks]

Take logs: $2x = \ln 7$ [1]; $x = (\ln 7)/2$ [1].

Self-assessment

Compare each credited step with your own answer. Fix the first reasoning or calculation error; then retry without looking at the solution.

Syllabus: Mathematics 9709, 2026–2027. <https://www.cambridgeinternational.org/Images/697427-2026-2027-syllabus.pdf>

Original FagusED teaching material. Question layout and mark presentation informed by the May/June 2026 papers supplied by the user. The practice question is newly written and is not a Cambridge past-paper question.